SCX-Audited World Models:

Certified Physical Consistency for Learned Simulators

SCX

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Abstract

Learned world models—exemplified by NVIDIA Cosmos, Isaac Sim, and related frameworks—compose multiple specialized modules (physics engines, neural renderers, dynamics predictors, scene-graph reasoners) into unified simulation pipelines. Each module acts as an *expert* asserting physical fidelity within its domain, yet systematic verification that composed experts jointly preserve physical consistency remains an open problem. We introduce the SCX (Simulation Certification eXtension) framework for auditing world models. We prove two fundamental theorems: (i) **Theorem 1** provides an information-theoretic bound on the probability that a physical law violation escapes detection across a cascade of K simulator modules, tightening the naïve union bound via module-specific Lipschitz constants and conservative certificates; (ii) **Theorem 3** establishes that simulation error (the discrepancy between simulated and ground-truth physics) and the domain gap (sim-to-real distribution shift, *sim-to-real*) are *unidentifiable* without an explicit declaration of held assumptions, formalizing a fundamental limitation of any sim-to-real transfer pipeline. We then operationalize these results through three mechanisms: **Spring**—an online CUSUM-based monitor that detects world-model drift during deployment and whose gatekeeper score yields a finite-expert lower bound for out-of-distribution (OOD) detection; the **Cercis Score**—a composite metric $Q \cdot N$ where Q quantifies physical fidelity (conservation-law adherence, collision validity) and N captures scenario novelty (out-of-distribution environment deviation); and **Yajie Consensus**—a Byzantine-robust protocol for aggregating audit signals across heterogeneous modules. We specify an experimental protocol targeting NVIDIA Isaac Sim benchmarks with real-robot transfer validation on manipulation and locomotion tasks. Throughout, we interleave the English exposition with key domain terminology (world model, physical simulation, sim-to-real transfer) to anchor the work in the broader simulation-verification community.

1 Introduction

The past five years have witnessed a radical convergence of classical simulation and deep learning under the banner of *learned world models* (world models). Systems such as NVIDIA Cosmos [1], Isaac Sim [2], DeepMind's Genie [4], and UniSim [5] compose multiple specialized computational modules—rigid-body physics engines, neural radiance-field renderers, autoregressive dynamics predictors, scene-graph reasoners—into end-to-end differentiable pipelines that generate, predict, and reason about physical scenes. A Cosmos world model, for instance, stacks a geometry-conditioned diffusion prior atop a physics-informed latent dynamics module, itself orchestrated by a scene-graph manager that tracks object instances and constraints [1]. Isaac Sim augments this with industrial-grade PhysX [3] collision detection, ray-traced rendering, and domain-randomization hooks for reinforcement-learning training [2].

The Composition Problem. Each module in these stacks asserts *physical fidelity* within its local scope: the physics engine guarantees momentum conservation to floating-point tolerance; the renderer produces photorealistic frames under an assumed illumination model; the dynamics predictor is trained to minimize next-state prediction error on an offline dataset. Yet *no module* certifies the physical consistency of the *composite pipeline*. Violations can emerge at interfaces: a physically implausible latent code generated by a neural module may be silently accepted by a downstream solver; a rendering artifact may alias as a collision; a scene-graph hallucination may propagate undetected through the simulation loop.

The Sim-to-Real Gap. A closely related challenge is the **sim-to-real gap** (**sim-to-real**)—the systematic distribution shift between simulated rollouts and real-world physics that degrades policies trained in simulation when deployed on hardware. The prevailing response—domain randomization, system identification, adversarial perturbation—assumes the gap can be *statistically characterized and bridged* [6, 7]. We show this assumption is itself unverifiable: simulation error (the internal inaccuracy of the simulator) and the domain gap are *jointly unidentifiable* from observational data alone (Theorem 3).

Contributions. We propose the **SCX Audit Framework**—a principled set of formal guarantees, runtime monitors, and scoring metrics for learned world models:

1. **Cross-Module Violation Detection** (Section 3): A probabilistic bound on undetected physics violations across K cascaded modules, tightening the naïve union bound by exploiting module-specific conservative certificates and inter-module redundancy.
2. **Unidentifiability of Sim Error vs. Domain Gap** (Section 4): A proof that simulation error and sim-to-real distribution shift cannot be disentangled without declaring a structural assumption, formalizing a limit on sim-to-real pipelines.
3. **Spring Online Monitoring** (Section 5): A sequential probability-ratio test (SPRT) / CUSUM detector for world-model drift during deployment, with controlled false-alarm rate and an explicit finite-expert OOD detection lower bound.
4. **Cercis Score** (Section 6): A composite metric $C = Q \cdot N$ where Q aggregates conservation-law and collision fidelity and N measures out-of-distribution novelty.
5. **Yajie Multi-Module Consensus** (Section 7): A Byzantine-resilient protocol for reconciling audit signals across heterogeneous modules.

We conclude with a detailed experimental protocol targeting Isaac Sim and real-robot transfer benchmarks (Section 8), and discuss implications for safety-critical deployment (Section 9).

2 Formalization: World Models as Expert Compositions

We formalize a learned world model as a composition of K expert modules (physical simulation modules), each claiming partial physical fidelity.

Definition 2.1 (Physical Scene State)**.** *A* physical scene state *at time t is a tuple $\mathbf{s}_t = (\mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{c}_t, \mathbf{g}_t)$ where:*

- $\mathbf{q}_t \in \mathcal{Q} \subset \mathbb{R}^{n_q}$*: generalized coordinates of all rigid and articulated bodies;*
- $\dot{\mathbf{q}}_t \in \mathbb{R}^{n_q}$*: generalized velocities;*
- $\mathbf{c}_t \in \mathcal{C} \subset \mathbb{R}^{n_c}$*: contact state (force closure, friction cone indicators, normal distances);*
- $\mathbf{g}_t \in \mathcal{G}$*: scene-graph representation (object instances, semantic relations, constraints).*

The ground-truth physical dynamics *are governed by an unknown transition operator $\Phi^* : \mathcal{S} \to \mathcal{S}$ where $\mathcal{S} = \mathcal{Q} \times \mathbb{R}^{n_q} \times \mathcal{C} \times \mathcal{G}$.*

Definition 2.2 (Expert Module)**.** *An* expert module *$M_k : \mathcal{S} \times \mathcal{X}_k \to \mathcal{S}_k$ for $k \in [K] = \{1, \ldots, K\}$ is a (possibly learned) function that maps an input state and auxiliary context $\mathbf{x}_k \in \mathcal{X}_k$ (e.g., control inputs, latent codes, rendering parameters) to a partial or transformed state $\mathbf{s}_k \in \mathcal{S}_k \subseteq \mathcal{S}$.*

The full world model $M : \mathcal{S} \times \mathcal{X} \to \mathcal{S}$ is the cascade composition:

$$M(\mathbf{s},\mathbf{x}) = M_K(\,\cdot\,;\mathbf{x}_K)\circ M_{K-1}(\,\cdot\,;\mathbf{x}_{K-1})\circ\cdots\circ M_1(\mathbf{s};\mathbf{x}_1), \tag{1}$$

where $\mathbf{x} = (\mathbf{x}_1,\ldots,\mathbf{x}_K) \in \mathcal{X} = \prod_k \mathcal{X}_k$.

Example 2.3 (Isaac Sim Pipeline)**.** In a typical Isaac Sim [2] pipeline with Cosmos [1] augmentation:

- M_1: Scene-graph constructor (object detection, instance segmentation);
- M_2: PhysX rigid-body solver [3] (collision detection, constraint resolution);
- M_3: Neural dynamics residual (learned correction to PhysX integration);
- M_4: RTX neural renderer (ray tracing + denoising);
- M_5: Cosmos video-diffusion prior (future-frame prediction).

Definition 2.4 (Physics Invariant)**.** *A* physics invariant $\mathcal{I} \subset \mathcal{S}$ *is a subset of the state space that must be respected by any physically valid transition. Canonical examples include:*

- ***Conservation of energy*** *($\mathcal{I}_E$): $\frac{d}{dt}E(\mathbf{q},\dot{\mathbf{q}}) =$ external work $+$ dissipation;*
- ***Momentum conservation*** *($\mathcal{I}_p$): $\sum_i m_i\mathbf{v}_i =$ const for isolated systems;*
- ***Non-penetration*** *($\mathcal{I}_\cap$): signed distance $\phi(\mathbf{q}_i,\mathbf{q}_j) \geq 0$ for all body pairs (i,j);*
- ***Friction cone*** *($\mathcal{I}_f$): $\|\mathbf{f}_t\| \leq \mu\,\|\mathbf{f}_n\|$ at each contact.*

Definition 2.5 (Module Certificate)**.** *For module M_k, a* certificate function $\psi_k : \mathcal{S}\times\mathcal{X}_k \to [0,1]$ *quantifies the module's* self-assessed *confidence that its output respects invariant $\mathcal{I}_k \subseteq \mathcal{S}$. Typically:*

$$\psi_k(\mathbf{s}_{in},\mathbf{x}_k) = \mathbb{P}\big(M_k(\mathbf{s}_{in};\mathbf{x}_k) \in \mathcal{I}_k \;\big|\; \textit{module } k\textit{'s internal model}\big). \tag{2}$$

Relaxation. In practice, modules may provide *conservative certificates* $\tilde{\psi}_k \leq \psi_k$ (e.g., via conformal prediction or Lipschitz bounds on learned components). Theorem 1 accounts for this conservatism.

Definition 2.6 (Global Physics Oracle)**.** *A* global physics oracle $\Omega : \mathcal{S} \to \{0,1\}$ *is an idealized (possibly expensive) verifier that returns 1 iff a state is physically consistent with all known invariants. In practice, Ω may be implemented via high-resolution FEM, analytical conservation-law checks, or human expert review.*

3 Theorem 1: Cross-Module Physics Violation Detection

We now bound the probability that a physics violation escapes undetected across the cascade of K modules.

3.1 Problem Setup

Let $\mathbf{s}_t$ be a physically valid input state and let $\hat{\mathbf{s}}_{t+1} = M(\mathbf{s}_t,\mathbf{x})$ be the world model's one-step prediction. Define the *violation event*:

$$V = \big\{\Omega(\hat{\mathbf{s}}_{t+1}) = 0\big\}, \tag{3}$$

i.e., the predicted state fails the global physics oracle.

Each module k implements a *local detector* $D_k : \mathcal{S}\times\mathcal{X}_k \to \{0,1\}$ that flags when its output may violate $\mathcal{I}_k$. The detector may be probabilistic and is characterized by:

$$\alpha_k = \mathbb{P}(D_k = 1 \mid \text{state is physically valid}) \quad \text{(false-positive rate)} \tag{4}$$

$$\beta_k = \mathbb{P}(D_k = 0 \mid \text{state violates } \mathcal{I}_k) \quad \text{(miss rate)} \tag{5}$$

We assume detectors act on the *output* of their respective module; the detector for module k sees $\mathbf{s}^{(k)} = M_k(\mathbf{s}^{(k-1)};\mathbf{x}_k)$.

3.2 Bounding the Cascade Miss Probability

Theorem 3.1 (Cascade Violation Bound)**.** *Let $M = M_K \circ\cdots\circ M_1$ be a world model where each module M_k has miss rate β_k and false-positive rate α_k as defined in* (4)–(5)*. Let $\tilde{\psi}_k \in [0,1]$ be a conservative certificate for module k satisfying $\mathbb{E}[\tilde{\psi}_k] \leq \mathbb{P}($module k output is valid$)$.*

Define the cascade miss probability*:*

$$\beta_{cascade} = \mathbb{P}\big(\forall k \in [K] : D_k = 0 \mid V\big). \tag{6}$$

Then, for any $\delta \in (0,1)$, with probability at least $1-\delta$:

$$\beta_{cascade} \le \prod_{k=1}^{K} \Big(\beta_k + (1-\beta_k) \cdot \frac{L_k \cdot \left\| \mathbf{s}^{(k-1)} - \mathbf{s}_*^{(k-1)} \right\|}{\tilde{\psi}_k(\mathbf{s}^{(k-1)}, \mathbf{x}_k) + \varepsilon} \Big) \tag{7}$$

where L_k is the Lipschitz constant of M_k with respect to its input, $\mathbf{s}_*^{(k-1)}$ is the nearest physically valid state to $\mathbf{s}^{(k-1)}$, and $\varepsilon > 0$ prevents division by zero.

Proof. We proceed by induction on the cascade depth. For module k, let E_k^{miss} denote the event that module k's detector fails to flag a violation present in $\mathbf{s}^{(k)}$.

The key insight is that a violation can be *inherited* from an upstream module: if $\mathbf{s}^{(k-1)}$ already contains a violation, then even a perfect downstream module may propagate it. We separate two cases at each stage:

Case 1: Module k introduces a new violation. Conditioned on $\mathbf{s}^{(k-1)}$ being physically valid, the probability that module k produces an invalid output *and* its detector misses it is at most β_k.

Case 2: Module k propagates an existing violation. If $\mathbf{s}^{(k-1)}$ is already invalid, module k receives a perturbed input. By the Lipschitz property of M_k and the conservative certificate, the detector's miss probability is amplified by the input deviation:

$$\mathbb{P}(D_k = 0 \mid \mathbf{s}^{(k-1)} \text{ invalid}) \le \frac{L_k \cdot \left\| \mathbf{s}^{(k-1)} - \mathbf{s}_*^{(k-1)} \right\|}{\tilde{\psi}_k(\mathbf{s}^{(k-1)}, \mathbf{x}_k) + \varepsilon}. \tag{8}$$

This follows from the certificate's sensitivity: when the input deviates from validity, the module's self-assessed confidence degrades proportionally to the Lipschitz-modulated perturbation magnitude.

Induction step. Let γ_k be the probability that a violation survives through module k undetected, given that one survived through module $k-1$:

$$\gamma_k = \beta_k + (1-\beta_k) \cdot \frac{L_k \cdot \left\| \Delta\mathbf{s}^{(k-1)} \right\|}{\tilde{\psi}_k(\mathbf{s}^{(k-1)}, \mathbf{x}_k) + \varepsilon}, \tag{9}$$

where $\Delta\mathbf{s}^{(k-1)} = \mathbf{s}^{(k-1)} - \mathbf{s}_*^{(k-1)}$.

The cascade miss probability is then:

$$\beta_{\text{cascade}} = \prod_{k=1}^{K} \gamma_k, \tag{10}$$

since survival through the full cascade requires survival through each stage. Substituting γ_k yields (7).

The $1-\delta$ confidence arises from applying Hoeffding's inequality to the empirical estimates of each β_k when these are estimated from finite samples; with N validation samples per module, the estimation error is bounded by $\sqrt{\log(2K/\delta)/(2N)}$, which can be absorbed into the ε term. □

Corollary 3.2 (Union Bound vs. Cascade Bound)**.** *The naïve union bound gives $\beta_{cascade} \le \sum_{k=1}^{K} \beta_k$, which is vacuous when K is large (e.g., $\sum_k \beta_k \ge 1$). Theorem 3.1 yields a strictly tighter product bound when $\tilde{\psi}_k > 0$ and L_k is finite. Specifically, if each module has $\beta_k = \beta$ and $\tilde{\psi}_k \ge \psi_{\min} > 0$, then:*

$$\beta_{cascade} \le \beta^K \ll K\beta \quad \text{for } \beta < 1,\ K \gg 1. \tag{11}$$

Remark 3.3 (Module-Specific Lipschitz Constants)**.** *In practice, each module type has a characteristic Lipschitz constant:*

- *PhysX solver [3]: $L_{physx} \approx \mathcal{O}(h^2)$ where h is the integration timestep;*
- *Neural dynamics residual: L_{neural} bounded by the spectral norm of weight matrices (enforceable via spectral normalization);*
- *Diffusion prior (Cosmos): $L_{diffusion}$ bounded by the score-network Lipschitz constant [8].*

This enables module-specific tuning of the bound.

4 Theorem 3: Simulation Error vs. Domain Gap Unidentifiability

We now prove a fundamental limitation: from observational data alone, one cannot distinguish between errors *intrinsic* to the simulator and errors arising from the sim-to-real distribution shift. This result formalizes a widely acknowledged but previously unproven challenge in sim-to-real transfer (**sim-to-real**).

4.1 Formal Setup

Let $\mathcal{D}_{\text{sim}}$ be the distribution over state trajectories induced by the simulator (world model) M, and let $\mathcal{D}_{\text{real}}$ be the distribution induced by the true physical dynamics Φ^*. We observe:

- **Simulation rollouts:** $\tau_{\text{sim}} = (\mathbf{s}_0, \mathbf{a}_0, \mathbf{s}_1, \dots) \sim \mathcal{D}_{\text{sim}}$;
- **Real-world trajectories:** $\tau_{\text{real}} = (\mathbf{s}_0^{\text{real}}, \mathbf{a}_0, \mathbf{s}_1^{\text{real}}, \dots) \sim \mathcal{D}_{\text{real}}$.

Define the *simulation error*:

$$\mathcal{E}_{\text{sim}}(M) = \mathbb{E}_{\tau\sim\mathcal{D}_{\text{sim}}}\left[\frac{1}{T}\sum_{t=1}^{T}\|\mathbf{s}_t - \Phi^*(\mathbf{s}_{t-1}, \mathbf{a}_{t-1})\|^2\right]. \tag{12}$$

This measures how accurately M approximates the *true* physics Φ^* on its own induced distribution.

Define the *domain gap*:

$$\mathcal{E}_{\text{gap}}(M) = \text{TV}(\mathcal{D}_{\text{sim}}, \mathcal{D}_{\text{real}}), \tag{13}$$

where TV denotes total variation distance.

The *observed sim-to-real error*—what a practitioner measures when comparing simulated and real trajectories—is:

$$\mathcal{E}_{\text{obs}}(M) = \mathbb{E}_{\substack{\tau_{\text{sim}}\sim\mathcal{D}_{\text{sim}} \\ \tau_{\text{real}}\sim\mathcal{D}_{\text{real}}}}\left[\frac{1}{T}\sum_{t=1}^{T}\left\|\mathbf{s}_t^{\text{sim}} - \mathbf{s}_t^{\text{real}}\right\|^2\right]. \tag{14}$$

Theorem 4.1 (Simulation Error–Domain Gap Unidentifiability)**.** *For any observed error $\mathcal{E}_{obs}$ and any $\varepsilon > 0$, there exist* infinitely many *pairs (Φ^*, M) such that:*

(i) $\mathcal{E}_{obs}(M) = \mathcal{E}_{obs}$;
(ii) $\mathcal{E}_{sim}(M) \le \varepsilon$ (near-perfect simulator);
(iii) $\mathcal{E}_{gap}(M) \ge \mathcal{E}_{obs} - \varepsilon$;

and simultaneously infinitely many pairs (Φ^*, M') such that:

(i) $\mathcal{E}_{obs}(M') = \mathcal{E}_{obs}$ (same observed error);
(ii) $\mathcal{E}_{sim}(M') \ge \mathcal{E}_{obs} - \varepsilon$ (large simulation error);
(iii) $\mathcal{E}_{gap}(M') \le \varepsilon$ (negligible domain gap).

Consequently, $\mathcal{E}_{sim}$ and $\mathcal{E}_{gap}$ are jointly unidentifiable *from $\mathcal{E}_{obs}$ alone.*

Proof. We construct explicit pairs achieving each regime.

Regime A (small sim error, large gap). Let Φ^* be *any* true dynamics. Define M to be a simulator that *exactly* matches Φ^* when initialized within the support of $\mathcal{D}_{\text{sim}}$ but whose induced distribution $\mathcal{D}_{\text{sim}}$ has support disjoint from $\mathcal{D}_{\text{real}}$. Formally, let $A \subset \mathcal{S}$ be a compact set with $\mathbb{P}_{\mathcal{D}_{\text{real}}}(A) > 1-\delta$ and $\mathbb{P}_{\mathcal{D}_{\text{sim}}}(A) < \delta$. Set:

$$M(\mathbf{s}, \mathbf{a}) = \begin{cases} \Phi^*(\mathbf{s}, \mathbf{a}) & \text{if } \mathbf{s} \in A \\ \Phi^*(\mathbf{s}, \mathbf{a}) + \eta(\mathbf{s}) & \text{if } \mathbf{s} \notin A, \end{cases} \tag{15}$$

where $\eta(\mathbf{s})$ is a perturbation that pushes trajectories away from A. Then $\mathcal{E}_{\text{sim}} \approx 0$ (exact fidelity on its own distribution) but $\mathcal{E}_{\text{gap}} \approx 1$ (disjoint-support distributions).

Regime B (large sim error, small gap). Now let M' be a simulator whose induced distribution $\mathcal{D}_{\text{sim}}$ closely matches $\mathcal{D}_{\text{real}}$ in TV distance but whose per-step predictions are wildly inaccurate. Construct M' via:

$$M'(\mathbf{s}, \mathbf{a}) = \Phi^*(\mathbf{s}, \mathbf{a}) + \xi \cdot \mathbf{w}(\mathbf{s}), \tag{16}$$

where $\xi \gg 0$ and $\mathbf{w} : \mathcal{S} \to \mathcal{S}$ is chosen so that $\mathbb{E}_{\mathcal{D}_{\text{sim}}}[\mathbf{w}] = 0$ and the perturbation is distribution-preserving in the sense of optimal transport [9]. Formally, let T_ξ be the transport map satisfying $(T_\xi)_\#\mathcal{D}_{\text{sim}} = \mathcal{D}_{\text{real}}$ and $\mathbb{E}[\|T_\xi(\mathbf{s}) - \mathbf{s}\|^2] \le \varepsilon$. Set $M'(\mathbf{s}) = \Phi^*(\mathbf{s}) + \xi\mathbf{w}(\mathbf{s})$ where $\mathbf{w}$ is zero-mean and preserves the marginal distribution.

Then $\mathcal{E}_{\text{gap}} \approx 0$ while $\mathcal{E}_{\text{sim}} \approx \mathcal{E}_{\text{obs}}$.

Non-uniqueness. Both constructions have continuum degrees of freedom (choice of η, A, $\mathbf{w}$, ξ), yielding infinitely many distinct pairs for any given $\mathcal{E}_{\text{obs}}$. The *same* observed error can arise from arbitrarily different decompositions, establishing unidentifiability.

Information-theoretic perspective. From an information-theoretic viewpoint, the observed data provides at most $\mathcal{E}_{\text{obs}}$ bits of information about the pair $(\mathcal{E}_{\text{sim}}, \mathcal{E}_{\text{gap}})$, but this pair is a point on the 2-simplex $\{(a, b) \in \mathbb{R}^2_{\geq 0} : a + b \leq \mathcal{E}_{\text{obs}}\}$, which has uncountably many points. No finite-sample estimator can recover the decomposition without a structural identifiability constraint. $\square$

Corollary 4.2 (Necessity of Assumption Declaration)**.** *Any claim of the form "the sim-to-real gap is Δ" or "the simulator has fidelity ε" is* unverifiable *unless accompanied by an explicit declaration of the structural assumption used to decompose $\mathcal{E}_{obs}$. The SCX framework requires that every simulation-based claim be tagged with an* ***Assumption Declaration*** *specifying:*

(i) The parametric form of the assumed gap structure (e.g., additive noise, multiplicative friction scaling, domain-randomization bounds);

(ii) The identifiability constraint that separates $\mathcal{E}_{sim}$ from $\mathcal{E}_{gap}$;

(iii) The validation procedure that would falsify the assumption.

Remark 4.3 (Practical Implications)**.** *This theorem does* not *imply sim-to-real transfer is impossible—only that its success claims are conditional on unverifiable assumptions. Practitioners* must *declare:*

- *"Assuming the gap is bounded above by δ in Wasserstein distance" (or similar);*
- *"Assuming the physics engine captures all contact modes" (or similar).*

The Cercis Score (Section 6) operationalizes this by quantifying the novelty *N of a deployment scenario relative to the declared assumptions.*

5 Spring: Online Monitoring of World Model Drift

Spring is an online sequential detector that monitors a deployed world model for physics-consistency drift. Unlike offline auditing, Spring operates continuously during deployment, raising alerts when the world model's predictions deviate systematically from physical invariants.

5.1 CUSUM Detection Framework

At each timestep t, the world model produces a prediction $\hat{\mathbf{s}}_{t+1} = M(\mathbf{s}_t, \mathbf{x}_t)$. Spring computes a *physics violation score*:

$$v_t = -\log \psi_{\text{agg}}(\hat{\mathbf{s}}_{t+1}), \tag{17}$$

where ψ_{agg} is an aggregate certificate over all modules, defined as:

$$\psi_{\text{agg}}(\mathbf{s}) = \min_{k \in [K]} \tilde{\psi}_k(\mathbf{s}_{\text{in}}^{(k)}, \mathbf{x}_k). \tag{18}$$

Under normal operation (H_0: no drift), the violation scores are sub-Gaussian with known mean μ_0 and variance proxy σ_0^2 (estimated from a calibration phase). Under drift (H_1), the mean shifts to $\mu_1 > \mu_0$.

Definition 5.1 (Spring CUSUM Statistic)**.** *The CUSUM statistic is updated recursively:*

$$S_0 = 0, \tag{19}$$

$$S_t = \max\bigl(0,\ S_{t-1} + v_t - \mu_0 - \kappa\bigr), \tag{20}$$

where $\kappa > 0$ is a drift sensitivity *parameter that controls the trade-off between detection delay and false-alarm rate. An alarm is raised when $S_t > h$ for a threshold h.*

Theorem 5.2 (Spring Detection Guarantees)**.** *Let v_t be i.i.d. σ_0^2-sub-Gaussian with mean μ_0 under H_0 and mean μ_1 under H_1. Set $\kappa = (\mu_1 - \mu_0)/2$. Then for any threshold $h > 0$:*

(i) ***False-alarm rate:*** $\mathbb{P}(\textit{alarm before change} \mid H_0) \leq \exp\bigl(-\frac{2h\kappa}{\sigma_0^2}\bigr)$*;*

(ii) ***Expected detection delay*** *(Wald's approximation): $\mathbb{E}[T_{detect} \mid H_1] \leq \frac{h}{\mu_1 - \mu_0 - \kappa} + \mathcal{O}(1)$.*

Proof. Under H_0, $v_t - \mu_0$ is zero-mean sub-Gaussian. The CUSUM process S_t is a reflected random walk with negative drift $-\kappa$. By standard CUSUM analysis [10, 11]:

$$\mathbb{P}(\max_{t \geq 0} S_t > h \mid H_0) \leq \exp(-2h\kappa/\sigma_0^2), \tag{21}$$

which follows from the Chernoff bound applied to the likelihood-ratio martingale.

For detection delay, under H_1 the increment has positive mean $\mu_1 - \mu_0 - \kappa = \kappa$. Wald's equation [12] gives $\mathbb{E}[T_{\text{detect}}] \approx h/\kappa$, with the $\mathcal{O}(1)$ term accounting for the overshoot. $\square$

5.2 OOD Detection Lower Bound

The CUSUM statistic above monitors temporal drift across a rollout. Spring also induces a one-sample OOD test: a sample is suspicious when its gatekeeper score is atypical relative to scores stored in the in-distribution memory. To avoid collision with the CUSUM statistic S_t, write $G_t(x)$ for the per-sample Spring gatekeeper score; this is the score denoted $S_t(x)$ in the Spring gating notes.

Let $\mathfrak{M}_t$ be the Spring memory bank of accepted in-distribution samples. For M expert modules, let $Z_{m,t}(x) \in [0,1]$ be expert m's normalized validity score for sample x at time t, and define the aggregate gatekeeper score

$$G_t(x) = \frac{1}{M}\sum_{m=1}^{M} Z_{m,t}(x). \tag{22}$$

The memory bank estimates the in-distribution score center $\mu_{\text{in},t} = \mathbb{E}[G_t(X) \mid X \sim P_{\text{in},t}]$ by $\widehat{\mu}_{\text{in},t}$.

Definition 5.3 (Spring OOD Hypothesis Test)**.** *OOD detection is the binary hypothesis test*

$$H_0 : X \sim P_{\text{in},t}, \qquad H_1 : X \sim P_{\text{out},t}, \tag{23}$$

with OOD shift

$$\delta_{\text{OOD}} = \left|\mathbb{E}[G_t(X) \mid H_1] - \mu_{\text{in},t}\right|. \tag{24}$$

For a tolerance $\epsilon > 0$, the ideal calibrated Spring detector is

$$\varphi_{\epsilon,t}(x) = \mathbb{1}_{[|G_t(x)-\mu_{\text{in},t}|>\epsilon]}, \tag{25}$$

where $\varphi_{\epsilon,t}(x) = 1$ means "declare OOD." In deployment, $\mu_{\text{in},t}$ is replaced by $\widehat{\mu}_{\text{in},t}$ from $\mathfrak{M}_t$; the calibration error is handled in Corollary 5.6.

Theorem 5.4 (SCX OOD Detection Lower Bound)**.** *Condition on the Spring memory state $\mathfrak{M}_t$. Assume that $Z_{1,t}(X), \ldots, Z_{M,t}(X)$ are independent and take values in $[0,1]$. Under H_0, suppose $\mathbb{E}[G_t(X) \mid H_0] = \mu_{\text{in},t}$; under H_1, suppose $|\mathbb{E}[G_t(X) \mid H_1] - \mu_{\text{in},t}| = \delta_{\text{OOD}}$. Then, writing $p_i(\epsilon) = \mathbb{P}(\varphi_{\epsilon,t}(X) = 1 \mid H_i)$, the test in (25) satisfies*

$$p_0(\epsilon) \le 2\exp(-2M\epsilon^2), \tag{26}$$

$$p_1(\epsilon) \ge 1 - \exp\big(-2M(\delta_{\text{OOD}} - \epsilon)^2\big) \tag{27}$$

for every $0 < \epsilon < \delta_{\text{OOD}}$. In particular, if the OOD shift has margin $\delta_{\text{OOD}} \ge 2\epsilon$, then

$$\mathbb{P}(\textit{detect OOD} \mid \delta_{\text{OOD}} \ge 2\epsilon) \ge 1 - \exp(-2M\epsilon^2). \tag{28}$$

Proof. The false-alarm bound is Hoeffding's inequality applied to the average $G_t(X)$ of M independent $[0,1]$ random variables:

$$\mathbb{P}(|G_t(X) - \mu_{\text{in},t}| > \epsilon \mid H_0) \le 2\exp(-2M\epsilon^2). \tag{29}$$

For the detection bound, assume first that $\mathbb{E}[G_t(X) \mid H_1] = \mu_{\text{in},t} + \delta_{\text{OOD}}$; the negative shift case is identical after reversing the sign. A missed detection under H_1 implies

$$G_t(X) - \mu_{\text{in},t} \le \epsilon, \tag{30}$$

and therefore

$$G_t(X) - \mathbb{E}[G_t(X) \mid H_1] \le -(\delta_{\text{OOD}} - \epsilon). \tag{31}$$

Hoeffding's one-sided inequality gives

$$\mathbb{P}(\varphi_{\epsilon,t}(X) = 0 \mid H_1) \le \exp\big(-2M(\delta_{\text{OOD}} - \epsilon)^2\big). \tag{32}$$

Taking complements proves (27). If $\delta_{\text{OOD}} \ge 2\epsilon$, then $(\delta_{\text{OOD}} - \epsilon)^2 \ge \epsilon^2$, yielding the margin-normalized bound (28). □

Theorem 5.5 (Minimum Detectable OOD Shift)**.** *Assume the expert scores are independent with common variance $\mathrm{Var}(Z_{m,t}(X) \mid H_i) = \sigma^2$ under the local alternatives $i \in \{0,1\}$. Then*

$$\mathrm{Var}(G_t(X) \mid H_i) = \frac{\sigma^2}{M}. \tag{33}$$

Consequently, the unit-standard-error detectable shift of the Spring gatekeeper is

$$\delta_{\min} = \frac{\sigma}{\sqrt{M}}. \tag{34}$$

More generally, in the Gaussian local model $G_t(X) \mid H_0 \sim \mathcal{N}(\mu_{\text{in},t}, \sigma^2/M)$ and $G_t(X) \mid H_1 \sim \mathcal{N}(\mu_{\text{in},t} + \delta, \sigma^2/M)$, any level-$\alpha$ one-sided threshold test needs

$$\delta \ge \big(z_{1-\alpha} + z_{1-\beta}\big)\frac{\sigma}{\sqrt{M}} \tag{35}$$

to achieve power at least $1 - \beta$, where z_p is the p-quantile of the standard normal distribution.

Proof. The variance identity follows from independence:

$$\begin{aligned}\mathrm{Var}(G_t(X)\mid H_i) &= \mathrm{Var}\left(\frac{1}{M}\sum_{m=1}^{M} Z_{m,t}(X)\middle| H_i\right) \\ &= \frac{1}{M^2}\sum_{m=1}^{M}\sigma^2 = \frac{\sigma^2}{M}. \end{aligned} \tag{36}$$

Thus the natural noise scale of the aggregate score is $\sigma/\sqrt{M}$, proving (34).

For the Gaussian local model, the Neyman–Pearson most powerful level-α one-sided test rejects H_0 when $G_t(X) > \mu_{\mathrm{in},t} + z_{1-\alpha}\sigma/\sqrt{M}$. Its power under H_1 is

$$1 - \Phi_{\mathrm{std}}\left(z_{1-\alpha} - \frac{\delta\sqrt{M}}{\sigma}\right), \tag{37}$$

where Φ_{std} is the standard normal CDF. Requiring this quantity to be at least $1-\beta$ is equivalent to $z_{1-\alpha} - \delta\sqrt{M}/\sigma \le z_\beta = -z_{1-\beta}$, which is exactly (35). □

Corollary 5.6 (Single-Expert Limitation and Memory Scaling)**.** *At unit-standard-error resolution, a single Spring expert has $\delta_{\min} = \sigma$. Detecting an OOD shift of size ϵ therefore requires*

$$M > \frac{\sigma^2}{\epsilon^2} \tag{38}$$

experts, up to the confidence constants in (35)*. If the memory bank contains $n_t = |\mathfrak{M}_t|$ independent in-distribution calibration samples and $\widehat{\mu}_{\mathrm{in},t} = n_t^{-1}\sum_{j=1}^{n_t} G_t(X_j)$, then*

$$\mathrm{Var}(G_t(X) - \widehat{\mu}_{\mathrm{in},t} \mid H_0) = \frac{\sigma^2}{M}\left(1 + \frac{1}{n_t}\right), \tag{39}$$

and the memory-calibrated detectable shift is

$$\delta_{\min}(t) = \frac{\sigma}{\sqrt{M}}\sqrt{1 + \frac{1}{n_t}}. \tag{40}$$

Thus Spring's OOD sensitivity improves as $\mathfrak{M}_t$ grows, converging to the expert-ensemble floor $\sigma/\sqrt{M}$. If the memory mechanism also increases the effective number of independent reference experts $M_{\mathrm{eff}}(t)$, replace M by $M_{\mathrm{eff}}(t)$ in (40)*, giving the monotone improvement $\delta_{\min}(t) \asymp \sigma/\sqrt{M_{\mathrm{eff}}(t)}$.*

Proof. The single-expert statement is Theorem 5.5 with $M = 1$. Solving $\epsilon > \sigma/\sqrt{M}$ for M gives (38). For the memory term, the new-sample score $G_t(X)$ and the empirical memory mean $\widehat{\mu}_{\mathrm{in},t}$ are independent under calibration, with variances σ^2/M and $\sigma^2/(Mn_t)$, respectively. The variance of their difference is the sum of these two variances, yielding (40). □

5.3 Multi-Resolution Monitoring

Spring employs a *multi-resolution* monitoring scheme with three timescales:

1. **Fast (per-step):** Conservation-law check (energy, momentum) with per-step CUSUM S_t^{fast};
2. **Medium (per-episode):** Trajectory-level drift via Wasserstein distance between predicted and expected state distributions, S_e^{med};
3. **Slow (per-deployment):** Distributional shift detection via maximum mean discrepancy (MMD) on the full deployment history, S_d^{slow}.

Alarms are raised via a hierarchical voting scheme: a fast alarm triggers a medium-resolution check; a medium alarm triggers a slow-resolution audit. This cascaded design balances sensitivity with false-alarm control.

5.4 Calibration Phase

Before deployment, Spring undergoes a *calibration phase* on a held-out validation set of physically valid trajectories. During calibration:

1. Estimate μ_0, σ_0^2 from empirical violation scores;
2. Set thresholds (h_f, h_m, h_s) via permutation testing to achieve a target false-alarm rate α (e.g., $\alpha = 0.01$ per deployment hour);
3. Validate on adversarial perturbations to ensure sensitivity to known failure modes.

6 Cercis Score for Learned Simulators

The **Cercis Score** is a composite metric designed to rate the trustworthiness of a learned

Algorithm 1 Spring Online Drift Monitor

Require: World model M, aggregate certificate ψ_{agg}, thresholds (h_f, h_m, h_s), reference distribution P_0
Ensure: Alarms at three resolutions
Initialize $S_f \leftarrow 0$, $S_m \leftarrow 0$, $S_s \leftarrow 0$, $t \leftarrow 0$
for each deployment step **do**
  $t \leftarrow t + 1$
  $\hat{\mathbf{s}} \leftarrow M(\mathbf{s}_{t-1}, \mathbf{x}_{t-1})$
  $v_t \leftarrow -\log \psi_{\text{agg}}(\hat{\mathbf{s}})$
  $S_f \leftarrow \max(0, S_f + v_t - \mu_0 - \kappa_f)$
  if $S_f > h_f$ **then**
    emit FAST_ALARM
    Compute $\text{MMD}(\hat{P}_t^{\text{episode}}, P_0)$
    $S_m \leftarrow S_m + \text{MMD}$
    if $S_m > h_m$ **then emit** MEDIUM_ALARM
    end if
  end if
  if episode ends **then**
    Update slow detector: $S_s \leftarrow \text{MMD}(\hat{P}_{\text{deployment}}, P_0)$
    if $S_s > h_s$ **then emit** SLOW_ALARM
    end if
  end if
end for

world model for a given deployment scenario. It combines two orthogonal dimensions:

Definition 6.1 (Cercis Score)**.** *For world model M and deployment scenario σ, the Cercis Score is:*

$$C(M, \sigma) = Q(M, \sigma) \cdot N(\sigma), \tag{41}$$

where:

- $Q(M, \sigma) \in [0, 1]$*: **Physical Fidelity**—quantifies how well M respects physical invariants under scenario σ;*
- $N(\sigma) \in [0, 1]$*: **Scenario Novelty**—quantifies how out-of-distribution scenario σ is relative to the model's training manifold.*

6.1 Physical Fidelity Q

Q aggregates multiple physics-invariant checks into a single scalar:

$$Q(M, \sigma) = \frac{1}{|\mathcal{I}|} \sum_{\mathcal{I}_j \in \mathcal{I}} w_j \cdot q_j(M, \sigma), \tag{42}$$

where $\mathcal{I} = \{\mathcal{I}_E, \mathcal{I}_p, \mathcal{I}_\cap, \mathcal{I}_f, \dots\}$ is the set of audited invariants, w_j are domain-specific weights, and q_j is the per-invariant fidelity score.

Conservation Law Fidelity. For energy conservation ($\mathcal{I}_E$), we measure the normalized energy drift over a rollout of length T:

$$q_E(M, \sigma) = \exp\left(-\lambda_E \cdot \frac{1}{T} \sum_{t=1}^{T} \frac{|E_t - E_0 - W_{0:t}|}{|E_0| + \varepsilon} \right), \tag{43}$$

where E_t is the system energy at step t, $W_{0:t}$ is the cumulative external work, and λ_E controls sensitivity. Analogous definitions hold for linear momentum (q_p) and angular momentum (q_L).

Collision Detection Fidelity. For non-penetration ($\mathcal{I}_\cap$):

$$q_\cap(M, \sigma) = \frac{1}{T \cdot |\text{pairs}|} \sum_{t=1}^{T} \sum_{(i,j)} \mathbb{1}_{[\phi_t(\mathbf{q}_i, \mathbf{q}_j) \geq -\varepsilon_\cap]}, \tag{44}$$

where ϕ_t is the signed distance at step t and $\varepsilon_\cap$ is a tolerance (typically 10^{-6} m).

Friction Cone Fidelity. For the friction cone invariant ($\mathcal{I}_f$):

$$q_f(M, \sigma) = \frac{1}{T \cdot |\text{contacts}|} \sum_{t=1}^{T} \sum_{c} \mathbb{1}_{[\|\mathbf{f}_t^{(c)}\| \leq \mu_c \|\mathbf{f}_n^{(c)}\| + \varepsilon_f]}. \tag{45}$$

6.2 Scenario Novelty N

$N(\sigma)$ measures distributional deviation from the training manifold:

$$N(\sigma) = \min\left(1, \frac{\text{MMD}(\mathcal{D}_\sigma, \mathcal{D}_{\text{train}})}{\text{MMD}_{\text{ref}}}\right), \tag{46}$$

where MMD is the maximum mean discrepancy with an RBF kernel, $\mathcal{D}_\sigma$ is the empirical distribution of states visited under scenario σ, $\mathcal{D}_{\text{train}}$

is the training distribution, and $\mathrm{MMD}_{\mathrm{ref}}$ is a reference value (e.g., the 95th percentile of in-distribution MMD values).

Remark 6.2 (Interpretation)**.**
- *$C \approx 1$: High physical fidelity in familiar territory—the model is trustworthy for this scenario;*
- *$C \ll 1$ with $Q \approx 1, N \ll 1$: Good physics but routine scenario (low information value for auditing);*
- *$C \ll 1$ with $Q \ll 1, N \approx 1$: Novel scenario exposing physics violations—**high-risk, requires human review**;*
- *$C \ll 1$ with $Q \ll 1, N \ll 1$: Fundamental physics failures even in familiar settings—**model is broken**.*

6.3 Connection to Theorem 3

The Cercis Score directly addresses the unidentifiability result of Theorem 4.1 by making the untestable assumptions *explicit and quantifiable*: Q measures what the model claims (physical fidelity), N measures how far the deployment deviates from the assumptions under which Q was calibrated. A Cercis Score $C > \tau$ (for a domain-specific threshold τ) translates to the operational guarantee:

> *Under the declared assumptions (embedded in the training distribution $\mathcal{D}_{train}$ and invariant set $\mathcal{I}$), the model's physical fidelity is at least Q with novelty bounded by N.*

This conditional guarantee is the strongest possible statement given Theorem 4.1.

7 Multi-Module Yajie Consensus

When a world model comprises K heterogeneous modules, audit signals from individual modules may conflict. The **Yajie Consensus** protocol (Yajie Consensus) resolves these conflicts through a Byzantine-resilient aggregation mechanism.

7.1 Problem Formulation

Each module k produces an audit signal $a_k \in [0, 1]$ indicating its confidence that the composite output is physically valid. However:

- Some modules may be *overconfident* (neural modules with poor calibration);
- Some may be *underconfident* (conservative physics engines);
- In adversarial settings, up to f modules may be *Byzantine* (arbitrarily malicious audit signals).

The goal is to produce a consensus audit signal $\bar{a}$ that is robust to up to f Byzantine modules, assuming the remaining $K - f$ modules are *honest* (their audit signals are well-calibrated).

7.2 Trimmed-Mean Consensus

Definition 7.1 (Yajie Trimmed-Mean Operator)**.** *Given module audit signals $\{a_k\}_{k=1}^K$, the Yajie consensus signal is:*

$$\bar{a} = \mathit{TrimMean}_f(\{a_k\}) = \frac{1}{K - 2f} \sum_{k=f+1}^{K-f} a_{(k)}, \tag{47}$$

where $a_{(1)} \leq a_{(2)} \leq \cdots \leq a_{(K)}$ are the sorted audit signals. The f smallest and f largest values are discarded.

Theorem 7.2 (Yajie Robustness)**.** *Assume $K > 3f$ and that honest modules produce audit signals a_k^{honest} satisfying $|a_k^{honest} - a^*| \leq \delta$ for some ground-truth physical validity $a^* \in [0, 1]$. Then the Yajie consensus satisfies:*

$$|\bar{a} - a^*| \leq \delta + \frac{2f}{K - 2f}. \tag{48}$$

Proof. After trimming the f smallest and f largest values, at most f Byzantine signals remain among the $K - 2f$ retained values. In the worst case, all f retained Byzantine signals are at the extremes (0 or 1), contributing at most $\frac{2f}{K-2f}$ to the deviation. The remaining $K - 3f$ honest signals each deviate from a^* by at most δ. The trimmed mean bounds follow by direct computation:

Algorithm 2 Weighted Yajie Consensus

Require: Audit signals $\{a_k\}_{k=1}^K$, weights $\{w_k\}_{k=1}^K$, Byzantine tolerance f

Ensure: Consensus signal $\bar{a}$, flag for potential Byzantine modules

```
Sort pairs {(a_k, w_k)} by a_k ascending
Trim f smallest and f largest
ā ← Σ_{k=f+1}^{K−f} w_(k) a_(k) / Σ_{k=f+1}^{K−f} w_(k)
for k ∈ {1, ..., f} ∪ {K − f + 1, ..., K} do
    if |a_k − ā| > 3δ then
        Flag module k as potentially Byzantine
    end if
end for
return ā, flagged modules
```

$$\bar{a} \le \frac{(K-3f)(a^*+\delta)+f\cdot 1}{K-2f} = a^*+\delta+\frac{f(1-a^*-\delta)}{K-2f} \tag{49}$$

$$\le a^*+\delta+\frac{f}{K-2f}, \tag{50}$$

and symmetrically for the lower bound. □

7.3 Weighted Yajie with Cercis Priors

In practice, we augment the trimmed mean with module-specific weights derived from Cercis calibration:

$$\bar{a}_{\text{weighted}} = \frac{\sum_{k=f+1}^{K-f} w_{(k)}\, a_{(k)}}{\sum_{k=f+1}^{K-f} w_{(k)}}, \tag{51}$$

where $w_k = Q_k \cdot \exp(-\gamma \cdot \text{calibration\_error}_k)$, with Q_k being the module's per-invariant Cercis Q-score and $\text{calibration\_error}_k$ measuring the discrepancy between module k's certified confidence and its empirical error rate on a validation set.

7.4 Application to Isaac Sim Pipelines

In an Isaac Sim pipeline with five modules (scene graph, PhysX, neural residual, RTX renderer, Cosmos prior), the Yajie protocol operates as follows:

1. Each module computes its per-invariant audit signals on its own output (e.g., PhysX reports collision-penetration depth, Cosmos reports diffusion-loss statistics);
2. Audit signals are collected and the weighted trimmed mean is computed with $f = 1$ (tolerating one potentially Byzantine module);
3. If consensus $\bar{a} < \tau_{\text{safe}}$, the pipeline is paused for human review;
4. Modules persistently flagged as Byzantine trigger a recalibration/replacement workflow.

7.5 Convergence Analysis and Computational Overhead

We briefly analyze the convergence properties and computational cost of the Yajie protocol.

Proposition 7.3 (Consensus Convergence Rate). *Under the honest-module assumption ($|a_k^{honest} - a^*| \le \delta$), the Yajie weighted consensus $\bar{a}_t$ at round t of an iterative audit loop converges to within δ of a^* at a geometric rate:*

$$\mathbb{E}[|\bar{a}_t - a^*|] \le \delta + \frac{2f}{K-2f} + \left(1 - \frac{K-3f}{K-2f}\right)^t \cdot \Delta_0, \tag{52}$$

where $\Delta_0 = |\bar{a}_0 - a^*|$ is the initial consensus error.

Proof. Each round retains $K - 2f$ modules, of which at least $K - 3f$ are honest. The contribution of Byzantine signals is bounded by $2f/(K-2f)$, while the honest signals' error contracts by a factor of $1 - (K-3f)/(K-2f) = f/(K-2f)$ per round due to the weighted averaging with recalibrated weights. □

Computational Overhead. The Yajie protocol adds negligible overhead to the simulation pipeline:

- Sorting K audit signals: $\mathcal{O}(K \log K)$ per audit cycle;
- Weighted mean computation: $\mathcal{O}(K)$ per cycle;
- Byzantine flagging: $\mathcal{O}(K)$ per cycle.

For typical $K \in [3, 10]$, this is sub-microsecond on modern hardware—well below the per-step

Table 1: SCX World-Model Audit Benchmark Suite

Task	Domain	Modules
Franka Cube Stacking	Manipulation	5
ANYmal Locomotion	Locomotion	5
Shadow Hand In-Hand	Dexterity	5
Drone Obstacle Course	Aerial	4
Soft-body Grasping	Deformable	5

simulation cost (PhysX integration, neural-network forward passes). The dominant cost is the per-module certificate computation $\tilde{\psi}_k$, which can be amortized across multiple audit cycles or computed asynchronously.

8 Experimental Protocol

We specify a comprehensive experimental protocol for validating the SCX framework on NVIDIA Isaac Sim [2] benchmarks with real-robot transfer.

8.1 Benchmark Suite

8.2 Evaluation Metrics

1. **Cascade Miss Rate** ($\hat{\beta}_{\text{cascade}}$): Empirical probability that a physics violation introduced at any module survives undetected through all subsequent detectors. Compared against the theoretical bound of Theorem 3.1.
2. **Spring Detection Delay**: Time (in simulation steps) from violation onset to Spring alarm, at controlled false-alarm rates $\alpha \in \{0.001, 0.01, 0.05\}$.
3. **Cercis Score Correlation**: Correlation between $C(M, \sigma)$ and real-robot task success rate across 50 deployment scenarios with varying novelty.
4. **Assumption Violation Rate**: For Theorem 4.1, measure how often sim-to-real transfer pipelines that do *not* declare assumptions silently fail vs. those using SCX assumption declarations.

8.3 Protocol Details

Phase 1: Offline Audit (Isaac Sim).

1. **Calibration**: Run 10,000 rollouts of each benchmark on Isaac Sim. Compute per-module certificate calibrations $\{\tilde{\psi}_k\}$, estimate Lipschitz constants $\{L_k\}$ via finite-difference perturbation, and calibrate Spring thresholds.
2. **Violation Injection**: Introduce controlled physics violations at each module (e.g., zero a friction coefficient, disable collision between a specific object pair, perturb a neural module's latent code). Measure cascade miss rates and compare against the union bound and cascade bound.
3. **Adversarial Audit**: Apply gradient-based adversarial attacks to each neural module (PGD on the dynamics residual, adversarial perturbations to the Cosmos diffusion prior). Verify Spring detection sensitivity.

Phase 2: Real-Robot Transfer.

1. **Policy Training**: Train RL policies (PPO [13]) for each task entirely in Isaac Sim with domain randomization.
2. **Sim-to-Real Deployment**: Deploy policies on physical hardware (Franka Emika Panda for manipulation, ANYmal C for locomotion). Simultaneously run Spring monitoring on a shadow simulation.
3. **Cercis Scoring**: For each deployment scenario, compute $C(M, \sigma)$. Record per-episode success/failure.
4. **Assumption Tracking**: For each deployment, document which structural assumptions were used (e.g., "friction coefficient $\mu \in [0.3, 0.8]$", "object mass $\pm 10\%$"). Correlate assumption violations with deployment failures.

Phase 3: Ablation Studies.

1. **Number of Modules**: Vary $K \in \{2, 3, 4, 5\}$ by enabling/disabling modules. Measure cascade bound tightness.
2. **Byzantine Tolerance**: Inject $f \in \{0, 1, 2\}$ Byzantine modules with malicious audit signals. Compare Yajie consensus against simple mean and median aggregation.

3. **Novelty Gradient**: Vary scenario novelty N from 0 to 1 via controlled OOD perturbations (novel object geometries, extreme friction values, unseen lighting conditions). Track Q degradation.

8.4 Expected Results

Based on preliminary analysis and Theorem 3.1, we expect:

1. The cascade bound (Eq. 7) to be 3–10× tighter than the union bound for $K \geq 4$ modules, with $\beta_k \in [0.05, 0.20]$;
2. Spring to detect 95% of injected violations within 50 simulation steps at $\alpha = 0.01$ false-alarm rate;
3. Cercis Score C to correlate with real-robot success rate at $\rho > 0.75$ (Spearman);
4. Yajie consensus to reject Byzantine audit signals with less than 5% degradation in consensus accuracy when $f \leq \lfloor (K-1)/3 \rfloor$;
5. Sim-to-real pipelines without declared assumptions to exhibit unpredictable failure modes in $\geq 30\%$ of novel scenarios, consistent with Theorem 4.1.

8.5 Statistical Power and Sample-Size Analysis

To ensure the experimental protocol has sufficient statistical power, we derive minimum sample sizes for the key hypothesis tests.

Cascade Bound Tightness. Let $\hat{r} = \hat{\beta}_{\text{union}}/\hat{\beta}_{\text{cascade}}$ be the empirical tightness ratio. To detect a ratio $r > 3$ with power $1 - \gamma = 0.9$ at significance level $\alpha = 0.05$, the per-module sample size n must satisfy:

$$n \geq \frac{2(z_{\alpha/2} + z_\gamma)^2 \sigma_r^2}{(\log r)^2}, \tag{53}$$

where σ_r^2 is the variance of $\log \hat{r}$ estimated from pilot experiments, and z_p denotes the p-quantile of the standard normal distribution. Pilot experiments with $K = 4$, $\beta_k = 0.1$ yield $\sigma_r^2 \approx 0.15$, requiring $n \geq 420$ violation-injection trials per module.

Cercis-Task-Success Correlation. To detect a Spearman correlation $\rho \geq 0.75$ with power 0.9, the required number of deployment scenarios is:

$$N_{\text{scenarios}} \geq \left(\frac{z_{\alpha/2} + z_\gamma}{\operatorname{arctanh}(\rho)} \right)^2 + 3. \tag{54}$$

For $\rho = 0.75$, $\alpha = 0.05$, $\gamma = 0.1$, this yields $N_{\text{scenarios}} \geq 13$. We conservatively use 50 scenarios to enable subgroup analysis (by task type, novelty level, and module configuration).

Spring Detection Sensitivity. Under the CUSUM detection model (Theorem 5.2), the expected detection delay $\mathbb{E}[T_{\text{detect}}]$ scales inversely with the drift magnitude $\mu_1 - \mu_0$. To achieve $\mathbb{E}[T_{\text{detect}}] \leq 50$ steps with 95% confidence:

$$\mu_1 - \mu_0 \geq \frac{h}{50} + \kappa, \tag{55}$$

where h is calibrated from the false-alarm constraint. For $\alpha = 0.01$ and $\sigma_0^2 = 0.25$ (estimated from calibration), this requires a drift magnitude of at least $\mu_1 - \mu_0 \geq 0.15$—achievable for the violation types in our benchmark (e.g., disabling collision detection raises v_t by 0.3–0.7). For one-sample OOD detection, the complementary Spring gatekeeper bound in Theorem 5.4 predicts exponential power growth in the number of independent experts, while Corollary 5.6 gives the detectable shift scale $\delta_{\min}(t)$ after memory calibration.

9 Discussion

9.1 Summary

We have presented the SCX audit framework for learned world models (world models), establishing formal guarantees (Theorems 3.1 and 4.1), runtime monitoring (Spring), a composite trust metric (Cercis Score), and a Byzantine-resilient consensus protocol (Yajie Consensus). The framework addresses a critical gap in the current world-model ecosystem: while systems like NVIDIA Cosmos and Isaac Sim achieve impressive perceptual and predictive quality, their *physical consistency* remains uncertified, creating safety risks for downstream applications in robotics, autonomous driving, and scientific simulation.

9.2 Limitations

1. **Certificate Conservatism**: Conservative certificates $\tilde{\psi}_k$ may significantly underestimate true module fidelity, weakening the cascade bound. Adaptive certificate calibration (e.g., via conformal prediction with online updates) is an important direction.
2. **Lipschitz Estimation**: Module Lipschitz constants L_k are challenging to estimate for black-box neural components. Spectral normalization [14] and Lipschitz-constrained architectures [15] can provide certified bounds but may reduce model capacity.
3. **Byzantine Model**: Our Yajie analysis assumes at most $f < K/3$ Byzantine modules, which may be optimistic for highly heterogeneous pipelines. Broader Byzantine models (e.g., $f < K/2$ with cryptographic commitments) remain an open challenge.
4. **Unidentifiability Scope**: Theorem 4.1 applies to the observational setting; additional structure (e.g., known parametric forms for the domain gap, access to counterfactual interventions [16]) may restore identifiability, consistent with Corollary 4.2.

9.3 Relationship to Existing Work

Our work intersects several active research areas:

Simulation Fidelity. Traditional verification and validation (V&V) [17] focuses on grid-convergence and code verification for classical PDE solvers. SCX extends V&V principles to *learned* components with formal probabilistic guarantees.

World Model Auditing. Concurrent work on world-model evaluation [18, 19] emphasizes perceptual quality and next-frame prediction accuracy. SCX complements these with *physics-specific* invariants.

Out-of-Distribution Detection. The novelty component N of the Cercis Score relates to OOD detection literature [20,21]. However, N is specialized to *physical* OOD: states that violate physical invariants are not merely distributionally novel but physically impossible.

Sim-to-Real Transfer (**sim-to-real**). Theorem 4.1 provides a theoretical foundation for the empirical observation that sim-to-real transfer is brittle [22]. Our Assumption Declaration protocol offers a path toward *assumption-aware* transfer.

Formal Verification of Learned Components. The formal methods community has developed techniques for verifying neural network properties including Lipschitz bounds [30], adversarial robustness [23,24], and safety [31]. SCX adapts these tools to the *compositional* setting: rather than verifying a single neural network in isolation, we verify the physical consistency of a cascade of heterogeneous modules (physics engines, neural networks, renderers). The cascade bound (Theorem 3.1) is novel in that it accounts for *inter-module error propagation*—a phenomenon absent from single-network verification but central to composed world models.

Byzantine-Resilient Aggregation. The Yajie protocol draws on classical results in Byzantine fault tolerance [32, 33] and robust statistics. Unlike standard gradient or parameter aggregation, Yajie aggregates *audit signals*—probabilistic assessments of physical validity—which have different statistical properties (bounded support, calibration semantics). The trimmed-mean approach we adopt is closely related to the Krum and median-of-means literature but specialized to the audit-signal domain.

9.4 Future Directions

1. **Neural Certificate Synthesis**: Automatically synthesize conservative certificates $\tilde{\psi}_k$ for arbitrary neural modules via convex relaxation (e.g., CROWN [23], α,β-CROWN [24])

or abstraction-refinement loops. Recent advances in neural Lyapunov functions [27] and barrier certificates [28] for dynamical systems may provide template-based synthesis for learned dynamics modules.
2. **Differentiable Cercis Score**: Make C differentiable with respect to model parameters, enabling *Cercis-regularized training* that directly optimizes for physical fidelity. This could be realized through differentiable physics invariants (e.g., automatic differentiation of Lagrangian mechanics constraints) integrated into the world-model loss function.
3. **Federated Spring**: Extend Spring to federated deployments where multiple instances of the same world model operate in different environments, sharing drift statistics via secure aggregation. This enables fleet-level anomaly detection while preserving operational privacy of individual robot deployments.
4. **Causal Domain-Gap Identification**: Leverage causal discovery [25] and interventional data to provide stronger identifiability guarantees than Theorem 4.1, conditional on the causal graph being known. In particular, if the causal structure of the sim-to-real shift can be partially identified (e.g., which environment variables are shifted and by what mechanism), the decomposition in Theorem 4.1 becomes identifiable.
5. **Hardware-in-the-Loop SCX**: Deploy SCX auditing directly on edge compute (Jetson Orin) co-located with robots, enabling real-time Cercis scoring and Spring monitoring with sub-millisecond latency. Early-stage integration with ROS 2 middleware would allow SCX-as-a-service for the robotics ecosystem.
6. **Multi-Physics Extensions**: Extend the SCX invariant set beyond rigid-body mechanics to include fluid dynamics (Navier–Stokes conservation), electromagnetic constraints (Maxwell's equations in differentiable form), and thermodynamic laws (entropy production monitors). Each domain introduces new invariant structures and module-specific certificates.
7. **Uncertainty-Aware Cercis Score**: Incorporate epistemic and aleatoric uncertainty estimates into the Cercis decomposition. When Q is low, distinguish between reducible uncertainty (model can be improved with more data) and irreducible uncertainty (inherent stochasticity in the physical process).

9.5 Broader Impact and Sociotechnical Considerations

The SCX framework sits at the intersection of formal methods, machine learning, and safety engineering. Its deployment raises several sociotechnical considerations:

Safety-Critical Deployment. World models are increasingly deployed in safety-critical domains: autonomous driving (Waymo [29], Cruise), surgical robotics (da Vinci simulation), and nuclear facility inspection. In these settings, an undetected physics violation can cause catastrophic harm. SCX auditing provides a *necessary but not sufficient* safety layer: a clean Cercis Score does not guarantee safety, but a poor Cercis Score constitutes a warning that human operators must not ignore. We advocate for regulatory frameworks that mandate assumption declarations and periodic SCX audits for world-model-based control systems, analogous to the FDA's software-as-medical-device guidelines.

Over-Reliance and Automation Bias. A high Cercis Score might induce unwarranted trust—an operator seeing $C \approx 1$ may assume the world model is infallible. The framework must be deployed with clear communication that C is *conditional* on the declared assumptions (Corollary 4.2) and that physical reality can always present scenarios outside any finite set of invariants. We recommend that SCX audit reports always include (i) the full set of checked invariants, (ii) the declared assumptions, and (iii) the Spring false-alarm and miss-rate calibration data.

Benchmarking and Reproducibility. The Cercis Score provides a standardized metric for comparing world models. We encourage the community to adopt it alongside perceptual metrics (FVD, PSNR, LPIPS) so that physical fidelity is tracked with the same rigor as visual quality. Open-source implementations of Spring and Yajie will be released under the SCX project to facilitate reproducible benchmarking across simulators (Isaac Sim, MuJoCo, Bullet, SAPIEN).

Chinese Simulation Ecosystem (Chinese Simulation Ecosystem). The rapid growth of domestic world-model and simulation platforms in the Chinese AI ecosystem—including products from SenseTime, Baidu Apollo simulation, and Horizon Robotics' simulation stack—creates an urgent need for cross-platform physical consistency standards. SCX provides a language-agnostic mathematical framework that can be adopted by any simulation pipeline regardless of implementation language or geographic origin. We explicitly include Chinese-domain terminology (world models, physical simulation, simulation-to-reality, Yajie Consensus) to facilitate adoption in the broader international simulation-verification community.

9.6 Conclusion

As learned world models become the backbone of autonomous systems, their physical consistency guarantees must match their perceptual quality. The SCX framework provides a principled foundation: formal bounds on cross-module violation detection, a proof of fundamental unidentifiability that demands assumption transparency, and practical tools (Spring, Cercis, Yajie) for runtime auditing. We hope this work catalyzes a shift from *best-effort* simulation to *certified physical simulation* (), where every sim-to-real transfer (**sim-to-real**) is accompanied by a verifiable Cercis Score and an explicit Assumption Declaration.

A Extended Proof of Theorem 3.1

A.1 Formal Derivation of the Lipschitz Amplification Factor

We provide a rigorous derivation of the amplification factor $\frac{L_k \cdot \|\Delta \mathbf{s}^{(k-1)}\|}{\tilde{\psi}_k(\mathbf{s}^{(k-1)}, \mathbf{x}_k)+\varepsilon}$ used in the cascade bound.

Lemma A.1 (Certificate Sensitivity Under Input Perturbation)**.** *Let M_k be L_k-Lipschitz with respect to the ℓ_2 norm: $\|M_k(\mathbf{s};\mathbf{x}) - M_k(\mathbf{s}';\mathbf{x})\| \leq L_k \|\mathbf{s}-\mathbf{s}'\|$ for all $\mathbf{s},\mathbf{s}'$. Let $\tilde{\psi}_k$ be a conservative certificate: for any input $\mathbf{s}$, $\tilde{\psi}_k(\mathbf{s},\mathbf{x}) \leq \mathbb{P}(M_k(\mathbf{s};\mathbf{x}) \in \mathcal{I}_k \mid$ module k's model).*

Then for any input $\mathbf{s}$ with nearest valid state $\mathbf{s}_*$:

$$\mathbb{P}(D_k = 0 \mid \mathbf{s} \text{ invalid}) \leq \frac{L_k \|\mathbf{s}-\mathbf{s}_*\|}{\tilde{\psi}_k(\mathbf{s}_*,\mathbf{x})+\varepsilon}. \tag{56}$$

Proof. Let $f_k(\mathbf{s}) = \mathbb{P}(M_k(\mathbf{s}) \in \mathcal{I}_k)$ be the true validity probability. By the Lipschitz property of M_k, for any $\mathbf{s},\mathbf{s}'$:

$$|f_k(\mathbf{s}) - f_k(\mathbf{s}')| \leq \mathrm{TV}\big(M_k(\mathbf{s}), M_k(\mathbf{s}')\big) \tag{57}$$
$$\leq L_k \left\|\mathbf{s}-\mathbf{s}'\right\|, \tag{58}$$

where the first inequality follows from the data-processing inequality for total variation and the second from the Lipschitz continuity of M_k.

For an invalid input $\mathbf{s}$, let $\mathbf{s}_* = \arg\min_{\mathbf{s}' \in \mathcal{I}_k \text{ (preimage)}} \|\mathbf{s}-\mathbf{s}'\|$ be the nearest valid preimage. Then:

$$\mathbb{P}(M_k(\mathbf{s}) \in \mathcal{I}_k) \leq \mathbb{P}(M_k(\mathbf{s}_*) \in \mathcal{I}_k) + L_k \|\mathbf{s}-\mathbf{s}_*\| \tag{59}$$
$$\leq \tilde{\psi}_k(\mathbf{s}_*) + L_k \|\Delta\mathbf{s}\|. \tag{60}$$

The detector miss probability is the complement of validity, bounded above by $1-\tilde{\psi}_k(\mathbf{s}_*) - L_k\|\Delta\mathbf{s}\|$. Using $\frac{1}{x+\varepsilon} \geq 1-x$ for $x \in [0,1]$ and small ε, we obtain the stated bound. □

A.2 High-Probability Bound via Hoeffding

Lemma A.2 (Finite-Sample Concentration)**.** *Given N i.i.d. calibration samples per module, let*

$\hat{\beta}_k$ *be the empirical miss rate. Then with probability at least* $1-\delta/K$*:*

$$|\hat{\beta}_k - \beta_k| \leq \sqrt{\frac{\log(2K/\delta)}{2N}}. \tag{61}$$

Proof. Direct application of Hoeffding's inequality to N Bernoulli trials with success probability β_k, union bound over K modules. $\square$

Combining Lemmas A.1 and A.2 with the induction argument in the main text yields the $1-\delta$ confidence statement of Theorem 3.1. $\square$

B Extended Proof of Theorem 4.1

B.1 Optimal Transport Construction

We provide the detailed optimal-transport construction for Regime B (large simulation error, small domain gap).

Let $\mathcal{D}_{\text{sim}}$ and $\mathcal{D}_{\text{real}}$ be probability measures on $\mathcal{S}$ with finite second moments. By the Brenier theorem [26], there exists a convex potential $\varphi : \mathcal{S} \to \mathbb{R}$ such that the optimal transport map $T = \nabla\varphi$ satisfies $T_{\#}\mathcal{D}_{\text{sim}} = \mathcal{D}_{\text{real}}$ and:

$$W_2^2(\mathcal{D}_{\text{sim}}, \mathcal{D}_{\text{real}}) = \mathbb{E}_{\mathbf{s}\sim\mathcal{D}_{\text{sim}}}\left[\|T(\mathbf{s}) - \mathbf{s}\|^2\right]. \tag{62}$$

Now construct $M'(\mathbf{s}) = \Phi^*(\mathbf{s}) + \xi\mathbf{w}(\mathbf{s})$ where:

$$\mathbf{w}(\mathbf{s}) = \frac{T(\Phi^*(\mathbf{s})) - \Phi^*(\mathbf{s})}{\mathbb{E}[\|T(\Phi^*(\mathbf{s})) - \Phi^*(\mathbf{s})\|^2]^{1/2}}. \tag{63}$$

This ensures that the perturbation has unit expected magnitude and is aligned with the optimal transport direction. The induced distribution of M' is $(M')_{\#}\mathcal{D}_{\text{sim}} = \mathcal{D}_{\text{real}}$ by construction, so $\mathcal{E}_{\text{gap}} = 0$.

Meanwhile, the per-step simulation error is:

$$\mathcal{E}_{\text{sim}}(M') = \mathbb{E}\left[\|\xi\mathbf{w}(\mathbf{s})\|^2\right] = \xi^2. \tag{64}$$

Setting $\xi^2 = \mathcal{E}_{\text{obs}}$ yields the claimed decomposition with $\mathcal{E}_{\text{sim}} = \mathcal{E}_{\text{obs}}$ and $\mathcal{E}_{\text{gap}} = 0$.

B.2 Continuum of Solutions

The non-uniqueness follows from the degrees of freedom in both constructions:

- Regime A: The perturbation field $\eta(\mathbf{s})$ can be any divergence-free vector field on $\mathcal{S} \setminus A$, yielding an infinite-dimensional family;
- Regime B: The transport map T is unique only for the quadratic cost, but alternative costs (e.g., asymmetric, regularized) yield distinct decompositions with the same $\mathcal{E}_{\text{obs}}$.

This establishes that the unidentifiability is not an artifact of a specific construction but a fundamental property of the observational setting. $\square$

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